

ATHENA-X

Pre-Market Report

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This report was generated by the ATHENA-X Institutional Report Engine. The engine is **read-only** and performs no calculations — every value originates from validated canonical databases or one of the seven DNA intelligence objects.

DNA Objects Consumed

DNA Object	Version	Confidence
Technical	dna-technical-v7.77	77.4%
Options	dna-options-v8.71	71.7%
Market	dna-market-v9.83	83.6%
Narrative	dna-narrative-v10.66	66.6%
Forecast	dna-forecast-v11.70	70.1%
Trade	dna-trade-v12.74	74.1%
Operations	dna-operations-v13.90	90.1%

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Executive Summary

Session: 2026-07-18

Generated: 7/18/2026, 00:40:56

Overall DNA Confidence: 76.25%

The market enters today's session with **SPDR S&P; 500 ETF at 583.61** (-0.24%) and **VIX at 18.42** (+0.09%).

Market Regime: trending

Trade Readiness: 72.00% — qualified setups available

Key Risk: Provider concentration on Polygon for market data

This report is composed from validated intelligence only. Every value can be traced back to a DNA object or canonical database record. No calculations are performed by the report engine.

Sources: market (dna-market-v9.83, conf 83.6%), trade (dna-trade-v12.74, conf 74.1%), operations (dna-operations-v13.90, conf 90.1%)

Market Overview

Symbol	Price	Change	Source
SPY	583.61	-0.24%	canonical.market_data (validated)
ES	5,807.69	-0.93%	canonical.market_data (validated)
SPX	5,840.23	-0.25%	canonical.market_data (validated)
QQQ	484.57	-0.70%	canonical.market_data (validated)
NQ	20,364.57	-0.42%	canonical.market_data (validated)
SOXX	232.01	+0.00%	canonical.market_data (validated)
VIX	18.42	+0.09%	canonical.market_data (validated)
VVIX	91.28	-0.89%	canonical.market_data (validated)
MOVE	102.50	-0.00%	canonical.market_data (validated)
TNX	4.27	-0.15%	canonical.market_data (validated)
DXY	105.07	+0.72%	canonical.market_data (validated)
Gold	2,391.55	-0.85%	canonical.market_data (validated)
Oil	81.23	-0.27%	canonical.market_data (validated)
Copper	4.17	-0.28%	canonical.market_data (validated)
USDJPY	155.41	-0.89%	canonical.market_data (validated)

Overnight Session:

- **Asia:** Nikkei 225 +0.84% — Strength in semiconductors; USDJPY volatility kept export names bid
- **Europe:** STOXX 600 -0.23% — Soft open; autos dragged while energy outperformed on supply concerns

Cross-Asset:

- **VIX:** 18.40 (+32.00%)
- **US 10Y:** 4.28 (+1.80%)
- **DXY:** 104.32 (-8.00%)
- **Gold:** 2,412.50
- **Oil:** 81.45
- **Copper:** 4.18
- **USDJPY:** 156.82

All prices sourced from the canonical market database (validated, cross-source checked).

Sources: market (dna-market-v9.83, conf 83.6%)

Technical Intelligence

Technical DNA Confidence: 77.41%

Freshness: 2,342.00ms

Contributors: 7 active

Trend Analysis

- **Primary Trend:** bullish
- **Timeframe Alignment:** multi-tf aligned
- **Structure:** HH/HL

Key Levels

- **Support:** 580.00
- **Resistance:** 595.00
- **VWAP:** 585.00

Momentum & Volume

- **RSI(14):** 55.0
- **MACD:** bullish cross
- **Volume Profile:** above average

Market Structure

- **Wyckoff Phase:** Phase D — markup
- **Chan Theory:** central zone, no signal

Liquidity

- **Bid Liquidity:** 12,400,000
- **Ask Liquidity:** 9,800,000
- **Liquidity Imbalance:** +12.00%

Sourced from Technical DNA (Stage 7). No recomputation performed.

Sources: technical (dna-technical-v7.77, conf 77.4%)

Options Intelligence

Options DNA Confidence: 71.74%
Freshness: 2,301.00ms
Contributors: 6 active

Dealer Positioning

- **Net Gamma:** +\$1.2B
- **Gamma Flip Level:** 580.00
- **Max Pain:** 582.00
- **Expected Move:** ±8.50 (1.50%)

Key Option Walls

- **Call Wall (Resistance):** 595.00
- **Put Wall (Support):** 578.00

Volatility

- **ATM IV:** 14.00%
- **IV Rank:** 42.00%
- **IV Percentile:** 38.00%
- **Term Structure:** contango

Risk Metrics

- **Theta Risk:** moderate — decay favors premium sellers
- **ODTE Risk:** elevated — gamma exposure above 1σ
- **Vanna Exposure:** neutral
- **Charm Exposure:** slight long bias

Sourced from Options DNA (Stage 8). No recomputation performed.

Sources: options (dna-options-v8.71, conf 71.7%)

Market Intelligence

Market DNA Confidence: 83.55%
Freshness: 2,977.00ms
Contributors: 5 active

Cross-Market Analysis

- **Equity-Vol Correlation:** -0.78
- **Equity-Bond Correlation:** 0.32
- **DXY-Equity Correlation:** -0.41

Sector Leadership

- **Semiconductors:** +1.20% — leading
- **Energy:** -0.40% — lagging
- **Financials:** +0.30% — neutral

Rotation

- **Rotation Signal:** defensive → cyclical
- **Breadth Thrust:** none detected

Breadth

- **Advance/Dcline:** 1.42
- **New Highs / New Lows:** 187 / 23
- **% Above 50-day MA:** 71.00%

Risk Score

- **Composite Risk:** 0.38 (0=low, 1=high)
- **Regime:** trending

Sourced from Market DNA (Stage 9). No recomputation performed.

Sources: market (dna-market-v9.83, conf 83.6%)

Narrative Intelligence

Narrative DNA Confidence: 66.65%
Freshness: 593.00ms
Contributors: 5 active

Story of the Day

Markets await CPI print; futures flat overnight with VIX elevated suggesting defensive positioning into the data.

Key Drivers

- CPI print at 8:30 AM ET
- Fed speakers throughout the day
- Earnings from 3 SPX components

Catalysts on the Radar

- **8:30 AM** — CPI release (high)
- **10:00 AM** — Wholesale inventories (low)
- **1:00 PM** — 10Y Treasury auction (medium)

News Feed

- **[HIGH]** CPI print scheduled for 8:30 AM ET — consensus 3.1% YoY — Benzinga Pro (7/18/2026, 00:30:56)
- **[MEDIUM]** Fed's Williams speaks at 11:00 AM ET on monetary policy outlook — Reuters (7/18/2026, 00:31:56)
- **[MEDIUM]** NVDA announces partnership with sovereign wealth fund — Bloomberg (7/18/2026, 00:33:56)
- **[MEDIUM]** Treasury announces 10Y auction at 1:00 PM ET — \$42B — US Treasury (7/18/2026, 00:35:56)
- **[MEDIUM]** ECB holds rates as expected; Lagarde hints at September cut — Reuters (7/18/2026, 00:36:56)
- **[MEDIUM]** China PMI misses expectations at 49.4 vs 49.6 consensus — Bloomberg (7/18/2026, 00:37:56)
- **[LOW]** Oil rises 1.2% on Middle East supply concerns — Bloomberg (7/18/2026, 00:38:56)
- **[LOW]** Initial jobless claims 218k vs 220k consensus — DOL (7/18/2026, 00:39:56)

Sentiment Trend

- **News Sentiment:** +12.00%
- **Social Sentiment:** +8.00%
- **Sentiment Divergence:** none — news and social aligned

Sourced from Narrative DNA (Stage 10) and canonical news database. No recomputation performed.

Sources: narrative (dna-narrative-v10.66, conf 66.6%)

Forecast Intelligence

Forecast DNA Confidence: 70.13%
Freshness: 266.00ms
Contributors: 5 active
Ensemble Models: 9 active

Horizon Forecasts

Horizon	Direction	Magnitude	Confidence
5 Minutes	bullish	+0.10%	62.00%
15 Minutes	bullish	+0.20%	65.00%
30 Minutes	neutral	+0.05%	58.00%
1 Hour	bullish	+0.30%	71.00%
End of Day	bullish	+0.60%	74.00%
Tomorrow	neutral	+0.10%	61.00%

Bull / Base / Bear Scenarios

- **Bull Case:** CPI cooler than expected; SPY tests 595 resistance with breakout to 600
- **Base Case:** CPI in line; SPY range-bound 582-590 into FOMC
- **Bear Case:** Hot CPI; SPY gaps below 578 put wall, targets 570

Calibration Status

- **Calibration Slope:** 0.970 (ideal = 1.0)
- **Directional Accuracy (30d):** 68.00%
- **MAE (30d):** 1.800 points

Model Contributions

- **LSTM-Price-v3:** weight 22.00%, contribution 71.00%
- **Transformer-Direction-v2:** weight 18.00%, contribution 65.00%
- **XGBoost-Vol-v1:** weight 14.00%, contribution 69.00%

Sourced from Forecast DNA (Stage 11). No recomputation performed — forecasts were generated upstream by the validated forecast ensemble.

Sources: forecast (dna-forecast-v11.70, conf 70.1%)

Trade Intelligence

Trade DNA Confidence: 74.15%
Freshness: 1,996.00ms
Contributors: 5 active

Trade Readiness

- **Readiness Score:** 72.00%
- **Qualified Setups:** 2 available
- **Checklist Status:** 7/8 criteria met — risk met, sizing pending

Active Trade Setup

- **Setup:** 0DTE Put Credit Spread
- **Direction:** neutral-bullish
- **Entry:** 585.00
- **Stop:** 578.00
- **Target:** 590.00
- **Risk:** \$1,250 / 1.2% portfolio
- **Reward:** \$750 / 0.7% portfolio
- **R/R Ratio:** 0.60
- **Probability:** 68.00%

Option Strategy

- **Strategy:** Sell 585P / Buy 580P (5-wide put credit spread)
- **Credit:** \$0.85 (\$85/contract)
- **Max Loss:** \$4.15 (\$415/contract)
- **Max Profit:** \$0.85 (\$85/contract)
- **Breakeven:** 584.15
- **Holding Time:** 0DTE — close at 4:00 PM ET

Risk Allocation

- **Position Size:** 2 contracts = \$830 max loss
- **Portfolio Heat:** 1.20%
- **Daily VaR:** 1.80%

Sourced from Trade DNA (Stage 12). No recomputation performed — all values produced by the validated decision intelligence layer.

Sources: trade (dna-trade-v12.74, conf 74.1%)

Risk Summary

Highest Risks

- **CPI surprise risk** (high) — size down 30% pre-print
- **Provider concentration on Polygon** (medium) — failover chain tested
- **0DTE gamma exposure** (medium) — avoid new entries 15min around CPI

Biggest Opportunities

- **IV crush post-CPI** (high) — window: 10:00 AM - 12:00 PM ET
- **Trend continuation if 590 breaks** (medium) — window: afternoon session
- **Defensive rotation** (medium) — window: anytime

Confidence Summary

DNA Object	Confidence	Status
Technical DNA	77.41%	✓ healthy
Options DNA	71.74%	⚠ degraded
Market DNA	83.55%	✓ healthy
Narrative DNA	66.65%	⚠ degraded
Forecast DNA	70.13%	⚠ degraded
Trade DNA	74.15%	⚠ degraded
Operations DNA	90.13%	✓ healthy

Warnings

- ⚠ Narrative DNA confidence below 65% — interpret news-driven signals with caution
- ⚠ VIX elevated above 20 — reduce position sizing by 20%
- ⚠ 0DTE gamma exposure 1.2σ above normal

Sourced from Operations DNA (Stage 13). No recomputation performed.

Sources: operations (dna-operations-v13.90, conf 90.1%)

Explainability

Every conclusion in this report answers four questions: **Why? Evidence? Supporting DNA? Confidence?**

Why is the trade readiness 72.00%?

Because the Trade DNA synthesis layer evaluated 7/8 criteria met — risk met, sizing pending against the active setup.

Evidence: 0DTE Put Credit Spread with entry 585.00, R/R 0.60.

Supporting DNA: Technical (confidence 77.41%) + Options (confidence 71.74%) + Forecast (confidence 70.13%).

Confidence: 74.15% — sourced from Trade DNA Stage 12.

Why is the market regime "trending"?

Because the Market DNA classified the current cross-asset state using sector leadership, breadth, and intermarket correlation.

Evidence: A/D ratio 1.42, 71.00% of issues above 50-day MA.

Supporting DNA: Market DNA (Stage 9) with 5 active contributors.

Confidence: 83.55%.

Why is the bull/base/bear forecast what it is?

Because the Forecast DNA ensemble of 9 active reached directional consensus on each horizon.

Evidence: Calibration slope 0.970 (ideal 1.0); 30-day directional accuracy 68.00%.

Supporting DNA: Forecast DNA (Stage 11) — all 9 model contributions preserved in the audit trail.

Confidence: 70.13%.

Audit Trail

- Every section in this report cites its source DNA object(s) by ID and version.
- The complete DNA snapshot is stored with the report (see audit metadata).
- No value in this report was computed by the report engine itself — all values originate from validated upstream intelligence.
- Report content hash: see audit metadata for the deterministic SHA-256 hash of this report's content.

Sources: trade (dna-trade-v12.74, conf 74.1%), market (dna-market-v9.83, conf 83.6%), forecast (dna-forecast-v11.70, conf 70.1%)

Audit Trail

Every stored report carries immutable audit metadata. The hash below is a deterministic SHA-256 of the report's structured content — identical inputs always produce identical hashes.

Schema Version	1.0
Generator Version	athena-x-report-engine-v15.0.0
Build Version	athx-15.0.0+sha.stage15
Forecast Version	fc-ensemble-v11.2
Content Hash (SHA-256)	a0e9e446d134b3507251f280c51d7ee358222eed9a307e14af0927889f61fb6b
DNA Versions	{ "technical": "dna-technical-v7.77", "options": "dna-options-v8.71", "market": "dna-market-v9.83", "narrative": "dna-narrative-v10.66", "forecast": "dna-forecast-v11.70", "trade": "dna-trade-v12.74", "operations": "dna-operations-v13.90" }
Workspace	default
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